準1級

第12回

英検準1級形式

リーディング練習

第12回　｜　部分演習（筆記全体の制限時間は流用しません）　｜　選択問題 31問

■ 出題構成・自己採点記録表

大問	問題番号	出題形式・内容	設問数	自己採点・点検
1	(1)〜(18)	短文の語句空所補充	18問	／18
2	(19)〜(24)	長文の語句空所補充	6問	／6
3	(25)〜(31)	長文の内容一致選択	7問	／7

■ 演習・印刷の手引き

1. 問題冊子には正答や解説を載せていません。解答用紙はA4横マークシート（1ページ）です。
2. 印刷時は「白黒・実際のサイズ（100%）」を選ぶと、マーク欄や英作文罫線が本番見本に近い実寸で出力されます。
3. 解き終わるまでは「解答・解説」を開かず、本番と同じ時間配分を意識して取り組んでください。
4. 同じ選択問題はアプリの「分野別練習」、英作文課題はアプリの「ライティング練習」でも学習・保存できます。

実施日：　　年　　月　　日　　開始〜終了：　　：　　〜　　：

氏名：＿＿＿＿＿＿＿＿＿＿＿＿＿＿＿＿　　選択問題 合計：　　／31 点

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1

次の(1)から(18)までの（ ）に入れるのに最も適切なものを1, 2, 3, 4の中から一つ選び、その番号のマーク欄をぬりつぶしなさい。

(1) Despite three rounds of mediation, talks collapsed over the union's ( ) refusal to accept two-tier pension benefits.

1 adamant **2** affable **3** amorphous **4** apposite

(2) Because neonatal sepsis progresses rapidly, pediatricians start broad-spectrum antibiotics at the first ( ) sign of infection.

1 indolent **2** incisive **3** incipient **4** insipid

(3) To keep invasive zebra mussels from clogging intake pipes, the water authority installed ultrasonic emitters that ( ) larval settlement.

1 defray **2** deter **3** delineate **4** deplore

(4) Although the startup's sleek pitch deck promised carbon-free cement, independent chemists dismissed its energy figures as ( ) wishful thinking.

1 staid **2** shrewd **3** scant **4** sheer

(5) Centuries of grazing by feral goats had stripped the volcanic island of shrubs, leaving steep slopes vulnerable to gully ( ) in winter rains.

1 erosion **2** erudition **3** equanimity **4** euphemism

(6) Far from clarifying authorship, ultraviolet imaging of the palimpsest revealed three overlapping scribes and deepened the textual ( ).

1 compendium **2** conundrum **3** concord **4** credence

1 短文の語句空所補充（続き）

(7) Under maritime salvage law, tugboat crews who risk their vessels to tow a drifting tanker off coastal reefs are entitled to ( ) compensation.

1 complacent **2** clandestine **3** commensurate **4** cursory

(8) Because rural clinics often lack refrigerated blood banks, biomedical engineers developed freeze-dried plasma that stays ( ) at room temperature.

1 volatile **2** venal **3** verbose **4** viable

(9) The Ombudsman faulted customs inspectors for applying baggage rules in a ( ) manner that varied wildly from shift to shift.

1 hardy **2** halcyon **3** haphazard **4** hallowed

(10) Rather than confront systemic safety lapses in the refinery, the board's report sought to ( ) a single night-shift technician.

1 scapegoat **2** scaffold **3** scavenge **4** scrutinize

(11) Even mild dehydration can ( ) a marathon runner's pace over the final ten kilometers on a humid morning.

1 bolster **2** blunt **3** burnish **4** buttress

(12) To prevent conflicts of interest, municipal procurement rules bar officials from accepting even ( ) gifts such as boxed fruit from contractors.

1 noxious **2** noisome **3** nomadic **4** nominal

1 短文の語句空所補充（続き）

(13) Once the dam's spillways were opened cautiously, reservoir pressure began to ( ) without flooding downstream orchards.

1 abate **2** abet **3** abjure **4** abrade

(14) Although both violinists played the Paganini caprice at breakneck speed, Clara's phrasing sounded effortless whereas Ken's felt ( ).

1 limpid **2** laudable **3** lenient **4** labored

(15) Only after archivists cross-referenced abbey tax rolls with tree-ring cores did a ( ) picture of the fourteenth-century drought emerge.

1 collateral **2** coercive **3** coherent **4** combustible

(16) Because bamboo culms reach full height in a single rainy season, foresters prize managed groves as a rapidly ( ) timber substitute.

1 reprehensible **2** renewable **3** revocable **4** rudimentary

(17) When unexpected bedrock fractures slowed tunnel boring, the contractor tried to ( ) lost weeks by running Sunday maintenance crews.

1 claw back **2** chalk up **3** brush off **4** bail out

(18) Before launching nationwide, the postal service decided to ( ) electric delivery tricycles in three hilly harbor wards.

1 talk over **2** turn down **3** try out **4** take in

2 A 次の英文を読み、その文意にそって(19)から(21)までの（ ）に入れるのに最も適切なものを1, 2, 3, 4の中から一つ選び、その番号のマーク欄をぬりつぶしなさい。

How Sand Batteries Store Renewable Winter Heat

Discussions of renewable energy storage almost always revolve around lithium-ion batteries wired into power grids. Chemical batteries excel at balancing hourly dips when clouds pass over solar farms, yet storing electricity across entire dark, freezing Nordic winters with lithium cells alone would be prohibitively expensive. In Finland and neighboring northern countries, however, more than half of winter energy demand goes not to running appliances or computers, but simply to keeping homes, schools, and factories warm.

That distinction inspired young Finnish engineers to build an astonishingly simple thermal store: an insulated steel silo packed with hundreds of tons of low-grade crushed soapstone or builders' sand rejected by glassmakers. Whenever fierce autumn gales spin wind turbines faster than households can use the electricity, cheap surplus power runs resistance heaters buried inside the silo, heating dry sand to around five hundred degrees Celsius. Because dry sand neither boils nor degrades at high temperatures—and thick mineral-wool insulation slows heat loss to a trickle—the core stays blistering hot for months. (***19***), operators blow air through heat-exchange pipes in January cold snaps to warm water for municipal district-heating pipes.

Converting sand heat back into electricity would waste roughly seventy percent of the energy, so sand silos are poor substitutes for grid batteries in sunny regions without winter heating needs. Used directly for space heating, process steam, or drying grain and timber, however, thermal efficiency exceeds eighty-five percent. (***20***), every ingredient—sand, steel, and resistive wire—is abundant, non-toxic, and free of scarce mining bottlenecks.

Matching storage chemistry to end use shows how low-tech materials can (***21***).

(19) **1** On the contrary **2** When needed **3** Otherwise **4** In vain

(20) **1** Reluctantly **2** Conversely **3** Instead of heat **4** Equally appealing

(21)

1 solve seasonal decarbonization challenges where fancy batteries fall short

2 power smartphones directly with handfuls of beach sand

3 keep tropical cities frozen throughout July and August

4 generate ninety-five percent electricity from lukewarm water

2 B 次の英文を読み、その文意にそって(22)から(24)までの（ ）に入れるのに最も適切なものを1, 2, 3, 4の中から一つ選び、その番号のマーク欄をぬりつぶしなさい。

Why Surge Pricing Provokes Moral Outrage

To a textbook microeconomist, raising prices dynamically when demand suddenly outstrips supply looks like plain common sense. If a torrential rainstorm hits a commuter rail station at six p.m. and three hundred drenched passengers compete for thirty nearby taxis, tripling fares does two useful things at once: travellers who can easily wait indoors stay put, while off-duty drivers two neighborhoods away gain a strong incentive to head toward the storm. Within half an hour, the queue clears.

In real life, however, dynamic "surge pricing" regularly triggers furious public backlash—especially when hardware stores raise snow-shovel prices after a blizzard or ride-hailing apps quadruple fares during transit breakdowns. Behavioral economists led by Daniel Kahneman, Jack Knetsch, and Richard Thaler traced this anger to an unwritten social norm they termed "reference-transaction fairness." Consumers accept that companies raise prices when wholesale costs rise—say, if cocoa beans double in price after crop disease. Exploiting a sudden spike in customer desperation without any matching rise in the seller's own costs, by contrast, violates deep intuitive rules of community reciprocity. (***22***), firms that cash in on a one-week emergency often forfeit years of hard-won customer loyalty.

Smart operators therefore look for ways to allocate scarce capacity without appearing predatory. Ride-hailing platforms cap multipliers during declared natural disasters while paying drivers bonuses out of company pockets; concert venues use randomized lotteries rather than pure auctions for affordable student seats. (***23***), utilities that pair peak-hour electricity surcharges with generous off-peak discounts frame dynamic rates as a reward for flexibility rather than a penalty.

Market design succeeds best when economists remember that prices carry (***24***).

(22) **1** On the contrary **2** Consequently **3** Otherwise **4** By coincidence

(23) **1** In vain **2** Conversely **3** Similarly **4** Despite discounts

(24)

1 guaranteed applause whenever snow shovels triple in price

2 zero emotional meaning for commuters caught in rainstorms

3 identical wholesale cocoa costs in every industry

4 moral signals about fairness alongside supply-and-demand math

3 A 次の英文の内容に関して、(25)から(27)までの質問に対する答えとして最も適切なもの、または文を完成させるのに最も適切なものを1, 2, 3, 4の中から一つ選びなさい。

How Ancient Roman Concrete Heals Its Own Cracks

Modern Portland-cement concrete—reinforced internally with steel rebar—underpins highways, seawalls, and skyscrapers across the globe, yet few twentieth-century bridges last more than fifty to a hundred years without costly patching. Once hairline cracks let moisture and road salt reach internal steel bars, rust expands and flakes surrounding concrete apart. By contrast, unreinforced Roman harbor piers pounded by Mediterranean waves for two millennia—and the vast coffered dome of the Pantheon in Rome—remain remarkably intact.

Architects long credited Roman durability almost entirely to volcanic ash, or pozzolana, quarried near Pozzuoli on the Bay of Naples and praised by the Roman engineer Vitruvius. Recent high-resolution X-ray spectroscopy by materials scientists in the United States, Italy, and Switzerland, however, uncovered a second secret hidden in plain sight: millimeter-wide white mineral lumps scattered throughout ancient mortar, known as "lime clasts." Researchers had previously dismissed those white flecks as sloppy mixing by careless laborers.

Chemical mapping proved instead that Roman builders deliberately "hot-mixed" reactive quicklime directly with volcanic ash and water at high temperatures. That exothermic reaction forged brittle calcium-rich pockets that act as a built-in repair reservoir. Whenever settling or earthquake tremors open a tiny crack in Roman concrete, rainwater seeping through the fissure dissolves calcium from those lime clasts; the saturated solution then recrystallizes as calcium carbonate, sealing the crack before it can widen—or reacts with volcanic ash to reinforce the surrounding matrix. Startup teams are now adapting hot-mixed lime chemistry to extend modern concrete lifespans and slash cement's heavy carbon footprint.

3A How Ancient Roman Concrete Heals Its Own Cracks　設問 (25)〜(27)

(25) Why does modern steel-reinforced concrete often deteriorate within several decades?

1 Portland cement dissolves instantly in fresh rainwater.

2 Modern builders mix too much volcanic ash from Naples into bridge piers.

3 Moisture and salt penetrate tiny cracks and rust internal steel bars, whose expansion breaks the concrete.

4 Steel rebar shrinks until bridges float away.

(26) How had researchers previously misinterpreted the white "lime clasts" visible inside Roman mortar?

1 They assumed the lumps were merely signs of careless, incomplete mixing.

2 They believed the white flecks were crushed Mediterranean pearls.

3 They thought Renaissance painters had added white spots for decoration.

4 They identified them as frozen salt crystals from winter roads.

(27) How do hot-mixed lime clasts enable Roman concrete to heal newly formed micro-cracks?

1 Internal steel magnets pull cracked concrete blocks back together.

2 Infiltrating water dissolves calcium from the clasts, which then recrystallizes to seal the fissure.

3 Volcanic heat keeps the Pantheon dome at two hundred degrees Celsius.

4 Tiny marine crabs cement broken pier stones with shells.

3 B 次の英文の内容に関して、(28)から(31)までの質問に対する答えとして最も適切なもの、または文を完成させるのに最も適切なものを1, 2, 3, 4の中から一つ選びなさい。

The Forgotten History of Pneumatic Mail Tubes

Long before fiber-optic cables or email servers linked metropolitan offices, nineteenth-century engineers built a physical high-speed messaging web beneath the streets of London, Paris, Berlin, Vienna, Prague, and New York: networks of cast-iron and brass pneumatic tubes. Powered by steam-driven air compressors and vacuum pumps, cylindrical leather-skirted canisters—stuffed with telegrams, stock orders, court filings, or urgent letters—whooshed around gentle underground bends at fifty kilometers per hour, bypassing horse-drawn gridlock above.

Pneumatic networks emerged in the 1850s and 1860s out of a paradoxical bottleneck in early electric telegraphy. Flashing Morse code fifty miles between Liverpool and London took seconds, yet delivering the transcribed paper slip the final two miles from a central telegraph exchange to a bank or newspaper desk by foot messenger could take over an hour. Connecting neighborhood post offices to central exchanges via pressurized tubes closed that "last-mile" gap. Paris eventually operated more than four hundred kilometers of municipal pipes; until 1984, Parisians could drop a blue carte pneumatique into special postboxes and expect delivery anywhere in the city within ninety minutes.

New York's twenty-seven-mile postal tube network carried up to ninety-five thousand letters per pipe per day at its peak, even whisking mail across the Brooklyn Bridge. Yet pneumatic grids proved rigid and maintenance-heavy as cities expanded outward. Moisture condensation jammed valves, digging up streets to find a stuck canister snarled traffic, and fixed pipe diameters could not scale to larger parcels. After mid-century budget disputes, New York shut its tubes in 1953, and Prague's flood-damaged network fell silent in 2002—leaving hospital lab-sample chutes as the last everyday heirs to a Victorian intranet.

3B The Forgotten History of Pneumatic Mail Tubes　設問 (28)〜(31)

(28) What specific bottleneck in early electric telegraphy spurred the invention of urban pneumatic tube networks?

1 Nineteenth-century banks banned electric telegraphs on weekdays.

2 Telegraph wires melted whenever operators sent more than ten words.

3 Steam locomotives refused to carry paper letters between Liverpool and London.

4 Morse signals crossed long distances in seconds, but carrying paper messages the final miles across congested city streets was slow.

(29) How effective was Paris's municipal pneumatic mail system for residents before it closed in 1984?

1 Special pneumatic cards could be delivered anywhere across the city within roughly ninety minutes.

2 Letters took three weeks to travel two blocks.

3 Only royal palace guards were allowed to mail canisters.

4 Canisters traveled exclusively on open-air rooftops.

(30) Which factors contributed to the eventual decline of municipal pneumatic mail grids in the twentieth century?

1 Canisters frequently exploded when crossing the Brooklyn Bridge.

2 A global shortage of leather skirts and brass after 1870.

3 Horse-drawn carriages became faster than fifty kilometers per hour.

4 Condensation issues, disruptive street excavations for stuck canisters, and fixed pipe sizes unable to handle larger packages.

(31) Where does the author note that pneumatic tube technology still survives in routine use today?

1 Inside transoceanic submarine internet cables.

2 Beneath every suburban driveway in New York.

3 In hospital chutes that transport laboratory samples.

4 Aboard commercial passenger airliners for luggage delivery.